



Vigneshwaran Neelakandan

Senior AI/Software Engineer
AI Engineering Services

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Senior Software Engineer with 4+ years of experience specializing in **AI and Deep Learning**, with strong expertise in **LLMs, RAG pipelines, AI agents, and domain-specific SLM fine-tuning**. Proficient in building **scalable, production-ready AI systems** using Python, with experience in **NL2SQL, schema linking, and reinforcement learning**.

WORK EXPERIENCE

Senior Software Engineer United Techno Info Systems

01/2025 - Present

Chennai, India

Projects/Products

- Architected and deployed an **on-premise LLM solution** using PEFT-based fine-tuning and quantization, enabling secure AI adoption for enterprise clients with strict data privacy requirements.
- Developed and fine-tuned **domain-specific Small Language Models (SLMs)** for high-accuracy industry applications.
- Architected and deployed **AI agents** to automate workflows, data validation, and customer support operations.
- Designed and deployed an **AI-powered website assistant using RAG and web search integration**, enabling real-time, context-aware responses and improving user engagement and lead generation.
- Designed and implemented **RAG pipelines with NER-based extraction** for intelligent document processing and knowledge retrieval.
- Optimized large-scale **data pipelines and analytics** using Python, SQL, and Snowflake for high-performance processing.
- Developed an **AI-powered data analytics platform (BI alternative)** using **LangChain**, enabling natural language queries (NL2SQL) across multiple data sources, generating real-time insights and dashboards via RAG and multi-LLM (OpenAI, Gemini, local SLM) integration.
- Applied **reinforcement learning techniques** for personalization and recommendation system optimization.
- Integrated **TruLens** for LLM evaluation, bias detection, and explainability in production AI systems.
- Built an **AI-driven test automation system using LLMs and LangChain**, generating test cases and executable scripts from URLs/Git code, with self-healing capabilities and automated reporting, reducing manual QA effort by **70%**.

SKILLS

Python LLM's GenAI Deep Learning
Transformers NLP RAG SLM
Langchain OpenCV NL2SQL NER
Azure AI foundry Azure App Services AWS
SQL REST API OpenAI APIs Gemini APIs

HONOR AWARDS

Excellence Award - 2024 (01/2024 - 12/2024)

United Techno Info Systems

Star Performer (01/2024 - 06/2024)

United Techno Info Systems

Star Performer (01/2023 - 03/2023)

United Techno Info Systems

SOFT SKILLS

Strong analytical and problem-solving skills for complex AI/DL systems
Effective communication with business stakeholders and clients
Experience in designing scalable AI solutions with system-level thinking
Ability to translate business requirements into technical implementations
Collaborative mindset working across engineering, product, and QA teams

INTERESTS

Exploring Generative AI tools and coding assistants (Claude, NotebookLM, Codex) for building intelligent workflows
Experimenting with AI agents, RAG systems, and automation use cases

WORK EXPERIENCE

○ Software Engineer United Techno Info Systems

05/2022 - 01/2025

Achievements/Tasks

- Developed a **clinical chatbot for hernia patient reports** using **Haystack and Macaw-based retrieval QA**, enabling doctors to query patient history and visit summaries via patient ID for faster clinical insights.
- Developed a **computer vision-based skin analysis system using OpenCV and Dlib**, leveraging AI to detect skin attributes and concerns, and recommend personalized products via vector search, achieving ~90% accuracy in skin issue identification.
- Developed a **FinOps AI agent** to automatically monitor Azure and AWS cost usage, generating daily and weekly AI-driven summaries with optimization insights for resource owners and leadership.
- Experience in AI driven solutions, data engineering, and intelligent automation.
- Proficient in Python, SQL, Snowflakes, Salesforce and DL libraries, with practical exposure to schema linking, reinforcement learning and Trulens for model evaluation.
- Managed **end-to-end deployment of AI applications on Azure App Services and IIS**, ensuring scalable, reliable, and secure production environments.

INTERESTS

Continuous learning and research in AI, Deep Learning, and emerging technologies

CERTIFICATES

NVIDIA GenAI Associate (11/2025 - 11/2027)

*Earned **NVIDIA Generative AI Associate Certification**, covering LLM fundamentals, prompt engineering, RAG architectures, and AI deployment best practices.*

PCEP Python Developer (12/2024 - Present)

*Earned **PCEP – Certified Entry-Level Python Programmer**, demonstrating foundational knowledge of Python syntax, data types, control flow, and basic programming concepts.*

EDUCATION

○ Vels University Vels Institute of Science Technology & Advanced Studies

07/2018 - 06/2022

Chennai, India

BE Computer Science and Engineering

- CGPA : 8.32